PAPER #89 ■ UQFF Master Equation: Complete Derivation

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Title: The UQFF Master Equation: Analytic Derivation and Implementation across 8 Calculator Architectures

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** validate*uqff*calculators.py, QCalc.UnifiedFieldSolver, 8 calculator classes **Index Slot:** ■1.12 UQFF Master Calculators, Paper #89

Abstract

The Unified Quantum Field Framework master equation F*U*Bi_i unifies gravity, electromagnetism, quantum effects, and vacuum dynamics in a single scalar. `validate_uqff_calculators.py` implements 8 specialized calculators (Base, Compressed, Superconductive, Triadic, Buoyant, MasterBuoyant, Resonant, Quadratic) derived from `QCalc.UnifiedFieldSolver`, each calling `.self_validate()` to confirm correctness. This paper presents the master equation derivation and documents the 8 derived forms with their domain of validity.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The UQFF Master Equation

The fundamental UQFF unified field evaluates as:

```
F_{U,Bi,i} = \int_{\mathcal{M}} \left[\sum_{k=1}^{4} U_{g_k}(r,t) + U_m(r) + U_{bi}(r,t)
+ \kappa(t) \cdot [{\rm SSq}] \right] dV
```

Where the integrand has 7 components:

Term	Symbol	Physics
Magnetic dipole	Ug1	BH/NS magnetosphere
Charge-reactivity	Ug2	Plasma-vacuum coupling
String rotation	Ug3	Rotational energy
Vacuum concentration	Ug4	BH-stellar boundary density
Magnetism	Um	Ambient magnetic field
Buoyancy force	U_bi	UQFF hydrostatic equilibrium
Calibration factor	?[SSq]	?=0.0005/day, [SSq]=0.57

2. Compact Form

The compact master equation (single symbol per body):

```
\mathbf{F}_U = \begin{pmatrix} U_{g1} \\ U_{g2} \\ U_{g3} \\ U_{g4} \end{pmatrix} \cdot
\vec{R}(r,t) + U_m + U_{bi} + \kappa[{\rm SSq}]
```

Where `\vec{R}` is the radial coupling vector.

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3. The 8 Calculator Architectures

3.1 UQFF_BaseCalculator

Domain: All astrophysical systems (general purpose)

```
F_{\rm Base}(r,t) = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m
```

Self-validation: checks all terms finite, Ug1■Ug4 > 0, Um > 0.

3.2 UQFF_CompressedCalculator

Domain: Galaxy clusters, cosmic web structures

```
F_{\rm Comp}(r,t) = g_{\rm MUGE}^{\rm Comp}(r) \cdot [1 + \delta_{\rm Ug}(r)]
```

Where `g_{\rm MUGE}^{\rm Comp}` uses 10-term compressed gravity (see Paper #90).

3.3 UQFF_SuperconductiveCalculator

Domain: Neutron stars, magnetars, near-horizon BH

```
F_{\rm SC}(r,t) = F_{\rm Base} \cdot [{\rm SCm}](r) = F_{\rm Base} \times 0.99
```

Self-validation: F*SC/F*Base ? (0.98, 1.00).

3.4 UQFF_TriadicCalculator

Domain: Triple systems and triple-resonance phenomena (3-body configurations)

```
F_{\rm Tri}(r,t) = \sum_{i=1}^{3} \left(F_{\rm Base}^{(i)} \cdot \cos\left(\frac{2\pi
(i-1)}{3}\right)\right)
```

Triadic phase: 120■ symmetry. Self-validation: F*Tri^(120■) = F*Tri^(0■) within 10?6.

3.5 UQFF_BuoyantCalculator

Domain: Atmospheric and plasma layers around compact objects

```
F_{\rm Buoy}(r,t) = F_{\rm Base}(r,t) + \rho_{\rm fluid}(r) g_{\rm eff}(r) \cdot [{\rm
UA}]
```

The [UA] = 0.0001 coupling ensures buoyancy correction is sub-dominant.

3.6 UQFF_MasterBuoyantCalculator

Domain: Full systems (BH + accretion disk + corona + jets)

```
F_{\rm MB}(r,t) = F_{\rm Buoy}(r,t) + \sum_k U_{g_k}^{\rm disk}(r_{\rm disk}) \cdot
A_{\rm AGN}(t)
```

Most complete single-body UQFF evaluation.

3.7 UQFF_ResonantCalculator

Domain: Resonance systems (magnetar multi-frequency, quasar QPOs)

```
F_{\rm Res}(r,t) = F_{\rm Base}(r,t) \cdot \left[1 + \sum_{n=1}^{5} a_n \cos(n\omega_0
t)\right]
```

5-frequency resonance: SuperFreq, QuantumFreq, AetherFreq, FluidFreq, ExpFreq (source27/28).

3.8 UQFF_QuadraticCalculator

Domain: Near-horizon corrections, Planck-scale physics

```
F_{\rm Quad}(r,t) = F_{\rm Base}(r,t) \left[1 + \beta_i
\left(\frac{r_P}{r}\right)^2\right]
```

With ■_i ■ 0.603 (calibrated constant, Batch 23). Post-GR quadratic corrections.

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4. UnifiedFieldSolver Integration

`QCalc.UnifiedFieldSolver` auto-selects calculator based on system parameters:

```
class UnifiedFieldSolver:
def select_calculator(self, system: dict) -> BaseCalculator:
if system['type'] == 'neutron_star':
return UQFF_SuperconductiveCalculator()
elif system['AGN_active']:
return UQFF_MasterBuoyantCalculator()
elif system['resonant']:
return UQFF_ResonantCalculator()
else:
return UQFF_BaseCalculator()
def self_validate(self) -> bool:
# Run all 8 calculators on standard test system
# Return True if all PASS
```

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5. Validation Results

All 8 calculators pass `self_validate()` on 5 standard systems (SgrA*, M87, Sun, NeutronStar, Magnetar):

Calculator	Self_Validate	Key Check
UQFF_BaseCalculator	PASS	All terms finite
UQFF_CompressedCalculator	PASS	MUGE compressed valid
UQFF_SuperconductiveCalculator	PASS	*FSC/F*Base ? (0.98,1.00)
UQFF_TriadicCalculator	PASS	120■ symmetry ■10?6
UQFF_BuoyantCalculator	PASS	Buoyancy < 1% correction
UQFF_MasterBuoyantCalculator	PASS	Full integration consistent

Calculator	Self_Validate	Key Check
UQFF_ResonantCalculator	PASS	All 5 frequencies finite
UQFF_QuadraticCalculator	PASS	■*i correction < 5% at r=r*Sch

Summary

The UQFF master equation admits 8 specializations covering all astrophysical regimes from smooth stellar systems (Base) to Planck-scale corrections (Quadratic). The `QCalc.UnifiedFieldSolver` seamlessly selects the appropriate calculator via system metadata, and all 8 pass self-validation.

Source: validateuqffcalculators.py | QCalc.UnifiedFieldSolver | all 8 self_validate() PASS

See also: PAPER_088 | Part of the Star-Magic UQFF Whitepaper Series.

UQFF computed: Canonical UQFF buoyancy parameter U*bi = ?■[SSq]■GM/r■ = 5.0e-4■0.57■6.67e-11■M/r■; for solar parameters: U*bi,Sun = 5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 m/s■.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
> the production physics constants and master equations to enable reproducibility
> against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-}$■ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`

Term	Description	Implementation
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
–Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.